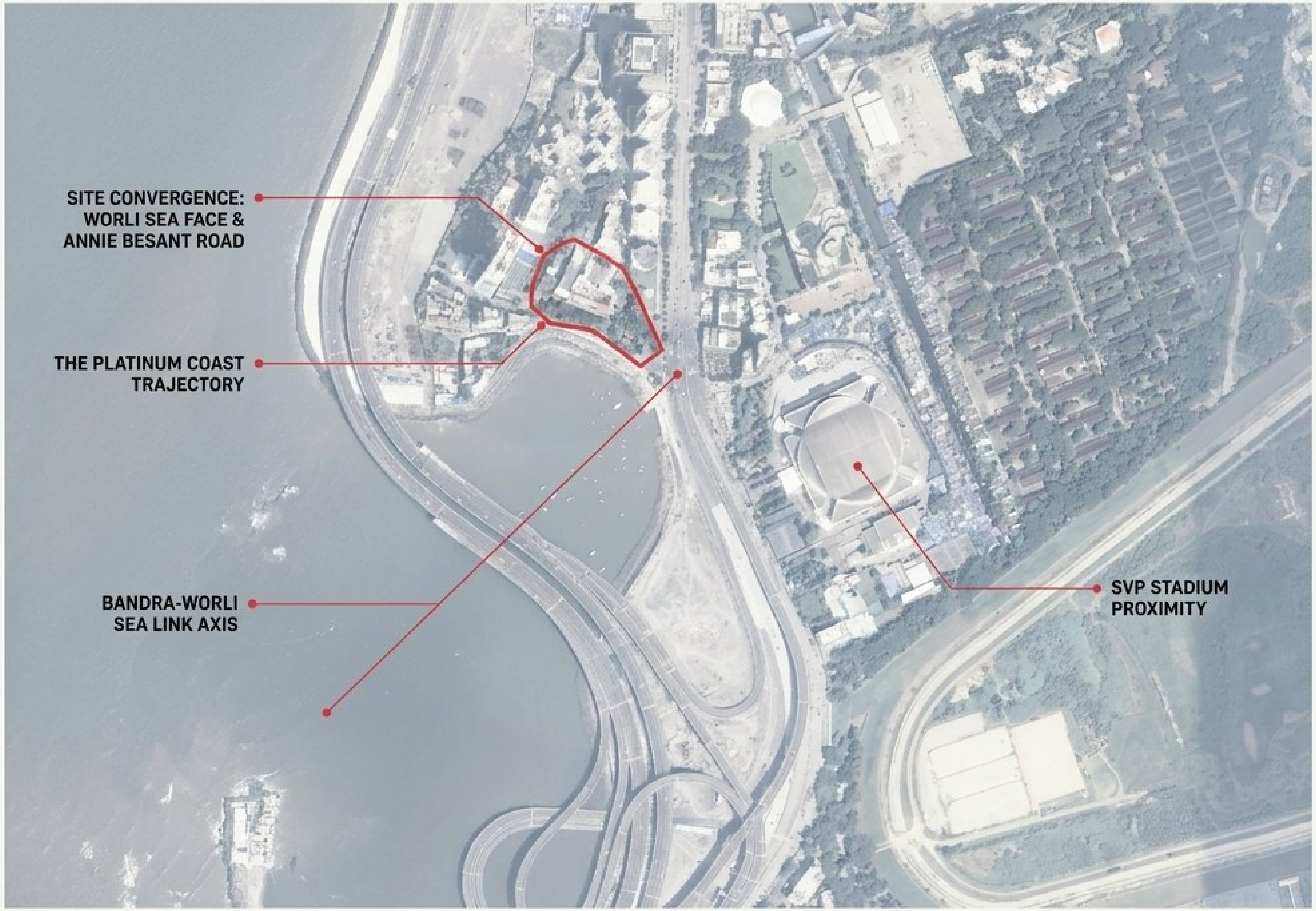
THE VERTICAL PARADIGM

Ultra-Luxury Redevelopment, High-Density Habitation, and Urban Integration

NSQF LEVEL 7 ADVANCED ARCHITECTURE DESIGN STUDIO



THE PLATINUM COAST CONTEXT

Worli has evolved from a landscape of low-rise cooperative societies into Mumbai's premier ultra-luxury destination.

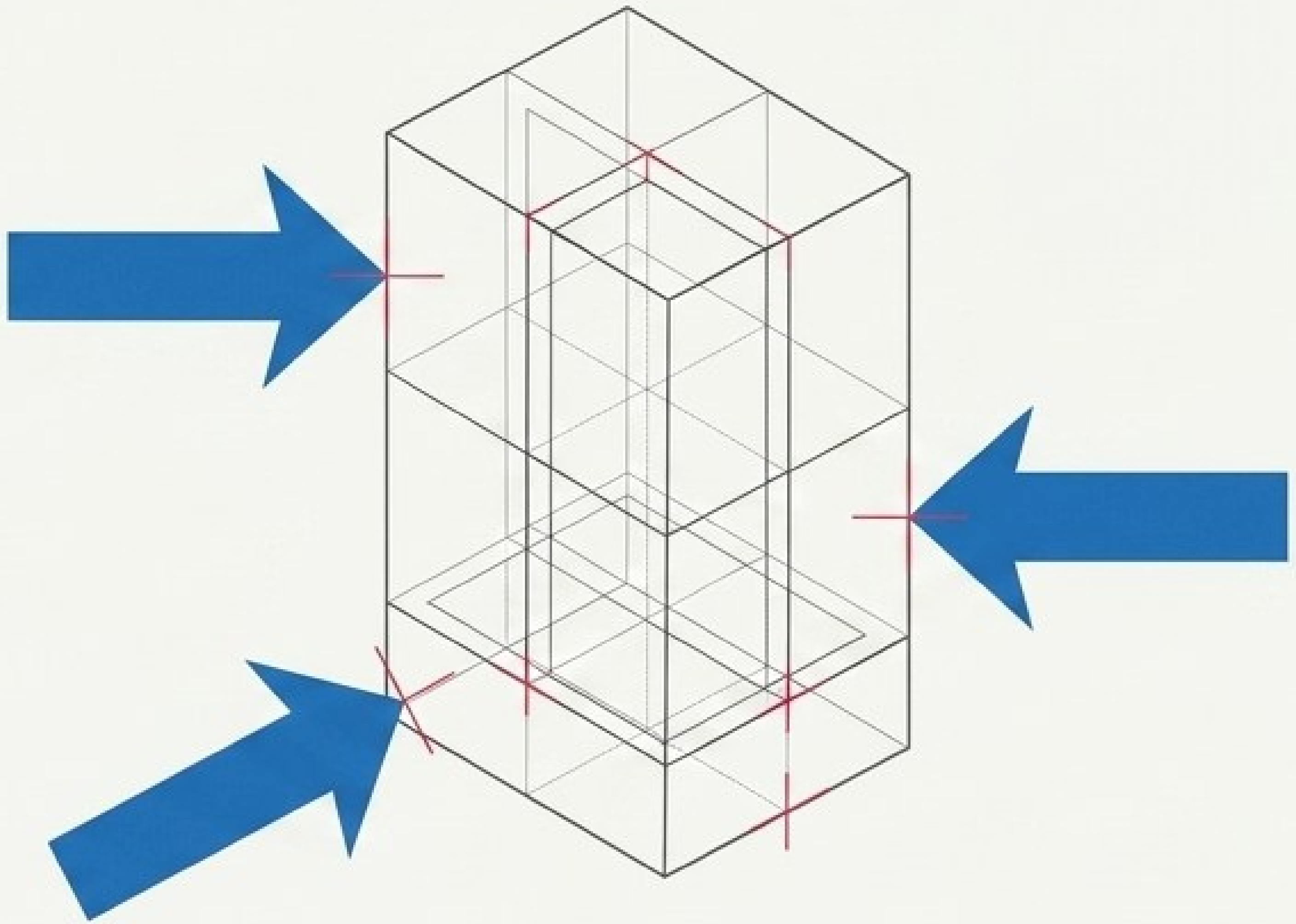
This brownfield redevelopment replaces an aging, sea-facing Cooperative Housing Society (Eden Hall) with architecture demanding global iconic status.

THE CORE COMPLEXITY: NAVIGATING URBAN FRICTION

Friction Vector Diagram

Constraint 1 (Legal/Environmental):

CRZ-II Coastal Regulation Zone norms. Extreme restrictions on ground coverage, basement depths, and setback lines.



Constraint 3 (Mathematical):

Operating within a rigid 4.0 FSI cap and DCR 2034 regulations, managing base FSI and 35% incentive area generation.

Constraint 2 (Aerodynamic/Structural):

The master tower must reach significant heights for unobstructed sea views, demanding aerodynamic massing to mitigate wind loads and advanced seismic systems.

Redevelopment in Mumbai is not merely an exercise in isolated luxury; it is a hyper-complex legal, structural, and environmental negotiation.

THE TRI-DEMOGRAPHIC DICHOTOMY

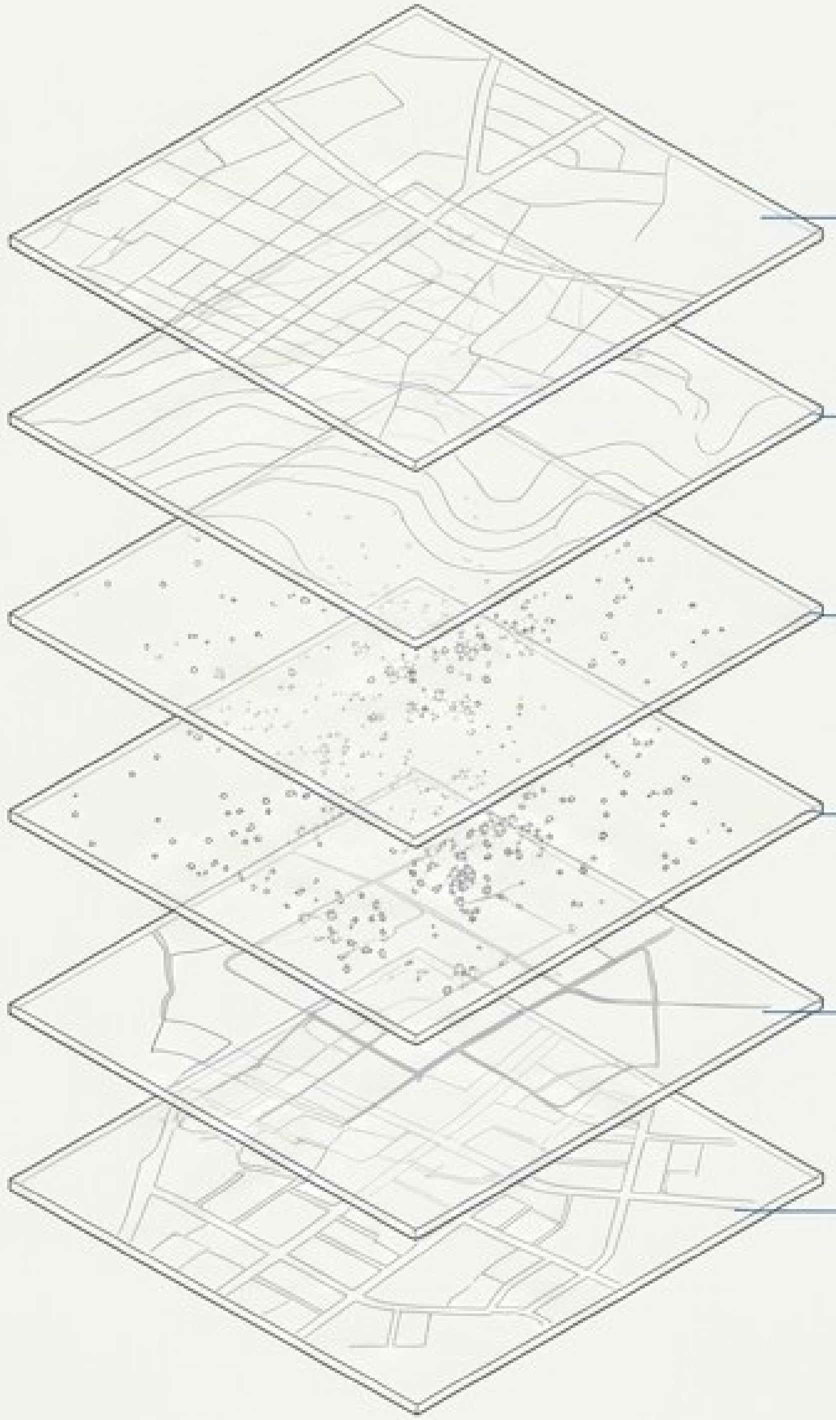
THE TRI-DEMOGRAPHIC ECOSYSTEM MATRIX		
THE CONTEXTUAL CIVIC PROGRAM	**THE REHABILITATION IMPERATIVE**	**THE ULTRA-LUXURY FREE SALE**
CORE MANDATE: Student-determined programmatic injection based on multi-layered urban analysis.	**CORE MANDATE:** Legal obligation to rehouse the existing Cooperative Housing Society families.	**CORE MANDATE:** Exclusive environment for elite urbanites, corporate leaders, and global investors.
SCALE: Exactly 15% of total FSI.	**SCALE:** 76 Identical Units.	**SCALE:** Maximum remaining FSI.
SPATIAL LOGIC: Public-facing urban integration, integrated into lower levels/podium.	**SPATIAL LOGIC:** Independent wing or demarcated tiers, featuring distinct lobbies and dedicated high-speed elevators.	**SPATIAL LOGIC:** Unobstructed sea views, grand arrival, exclusive lifestyle ecosystem.

SITE DATA & ARCHITECTURAL BOUNDS

8,248 sq.m	**PLOT SIZE** (Approx. 2.03 Acres / ~88,781 sq.ft)
4.0	**MAX PERMISSIBLE FSI**
355,124 sq.ft	**TOTAL PERMISSIBLE BUILT-UP AREA (BUA)** (excluding fungible areas, balconies, podium parking)
280 Meters	**AVIATION CLEARANCE HEIGHT RESTRICTION**

Design must successfully translate Mumbai's DCR 2034 into volumetric massing models.

The 6-Layer Urban Analysis



PERCEPTUAL: Lynchian analysis (nodes, edges, paths, districts), micro-climate sensory experiences.

VISUAL: Townscape, serial vision, sightlines, skyline profiles.

TEMPORAL: Diurnal rhythms (day/night economies), seasonal shifts (monsoons), historical evolution.

SOCIAL: Human behavior, demographic clustering, informal economic appropriation, inclusion/exclusion.

FUNCTIONAL: Land use, logistics, economic networks, commercial vs. residential distribution.

MORPHOLOGY: Physical grain, building typologies, density gradients, street hierarchies.

Outputs the 53,268 sq.ft Contextual Urban Program.

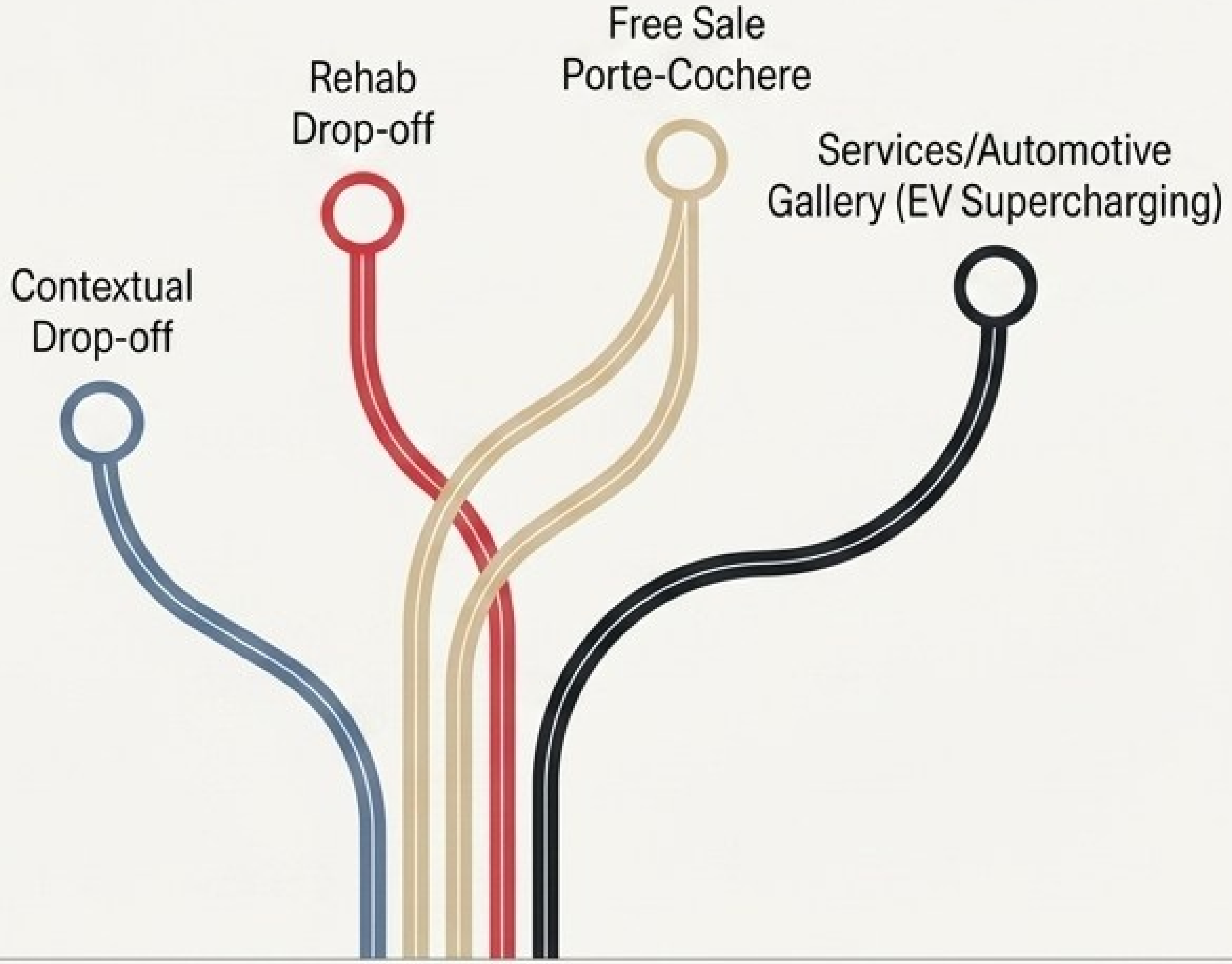
Mathematical Resolution: The Elite Market Ruleset

TOTAL AVAILABLE FREE SALE AREA: 199,256 SQ.FT.

THE FREE SALE INVENTORY DIAGNOSTIC TABLE					
1	**TYPOLOGY 1: THE TOWNHOUSES**	**Rule:** Exactly 10 units.	**Specs:** 3,000 sq.ft each.	**Integration:** Podium level with private garden interfaces.	**AREA CONSUMED: 30,000 SQ.FT.**
			REMAINING AREA: 169,256 SQ.FT		
2	**TYPOLOGY 2: 3BHK TOWER RESIDENCES**	**Rule:** Size range of 1,400 to 1,800 sq.ft.	**Ratio Rule:** Must constitute exactly exactly 60% of total tower units.		
3	**TYPOLOGY 3: 4BHK TOWER RESIDENCES**	**Rule:** Size range of 1,800 to 2,500 sq.ft.	**Ratio Rule:** Must constitute exactly 40% **40%** of total tower units.		

TASK: STUDENTS MUST CALCULATE THE EXACT UNIT COUNT AND PRECISE SQUARE FOOTAGES TO PERFECTLY CONSUME THE 169,256 SQ.FT LIMIT WITHOUT REMAINDER.

The Ground Plane & Lifestyle Ecosystem



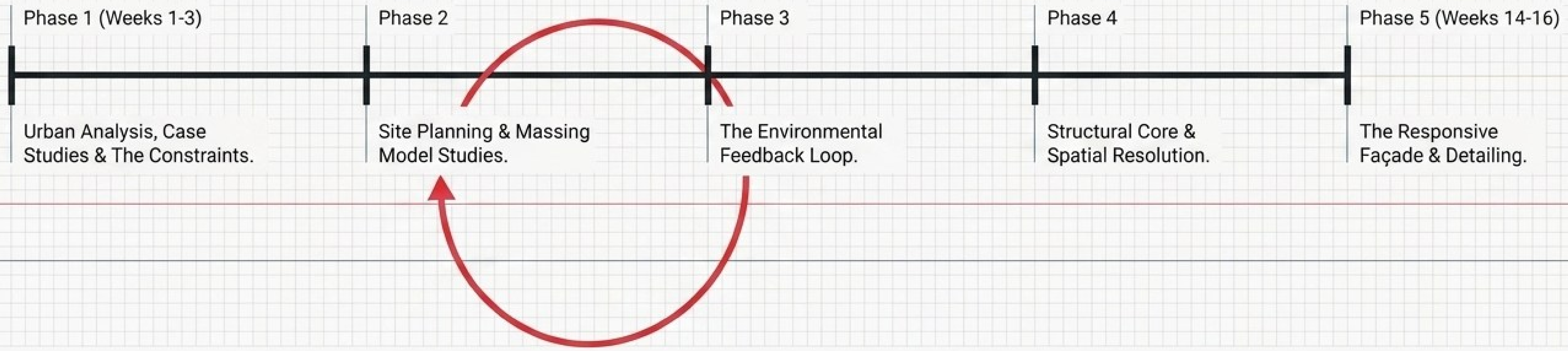
The Grand Luxury Clubhouse

PROGRAM:

Sprawling podium deck exclusively for Free Sale units. 50m Olympic-length infinity pool, wellness spa, squash courts, fine-dining banquet hall.

Execution Methodology: The Iterative Loop

A rigorous studio workflow integrating legal boundaries, evidence-based simulation, and structural reality.



Phases 1 & 2: Constraints to Volumes

PANEL A: EXPERT INTEGRATION

SITE VISITS & HIGH-RISE EXECUTION:
Ar. Mandar Ghosalkar & Ar. Bina Bhatia.

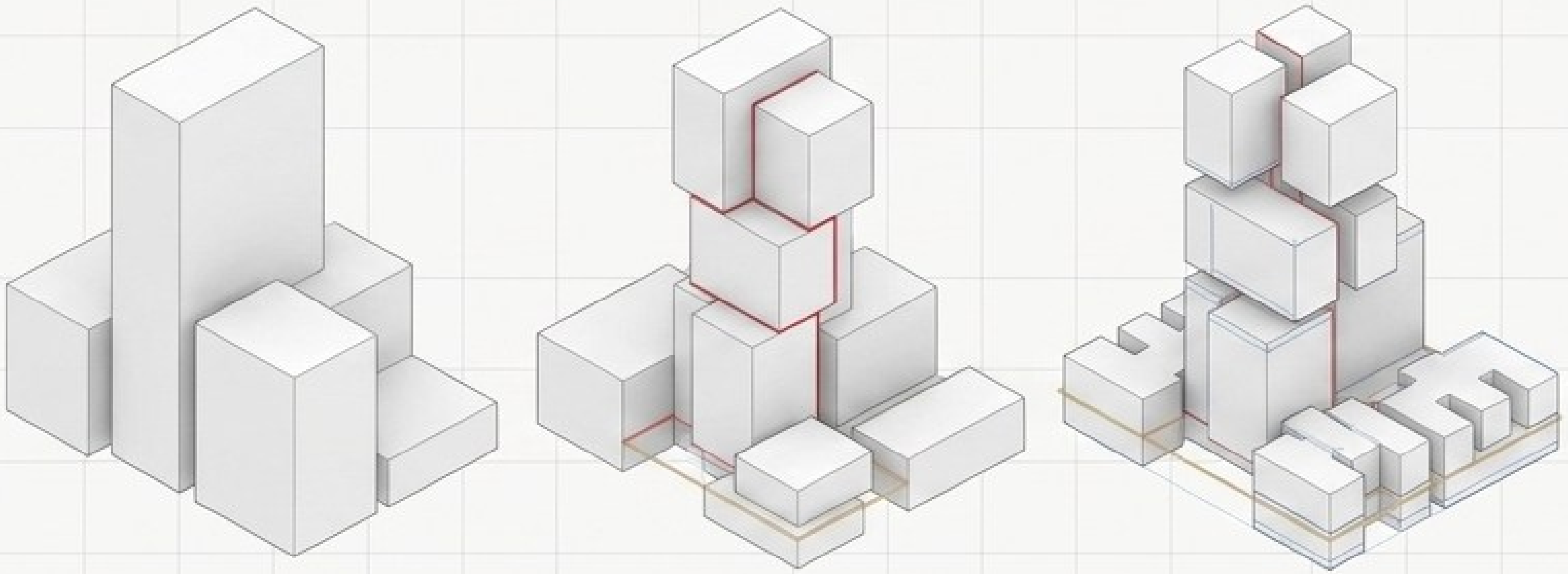
DECODING MUMBAI BYE-LAWS:
Kiran Rao (DCR 2034 & CRZ regulations).

HIGH-RISE STRUCTURAL DESIGN:
CTBUH Representative (core design, outrigger trusses, wind-load mitigation).

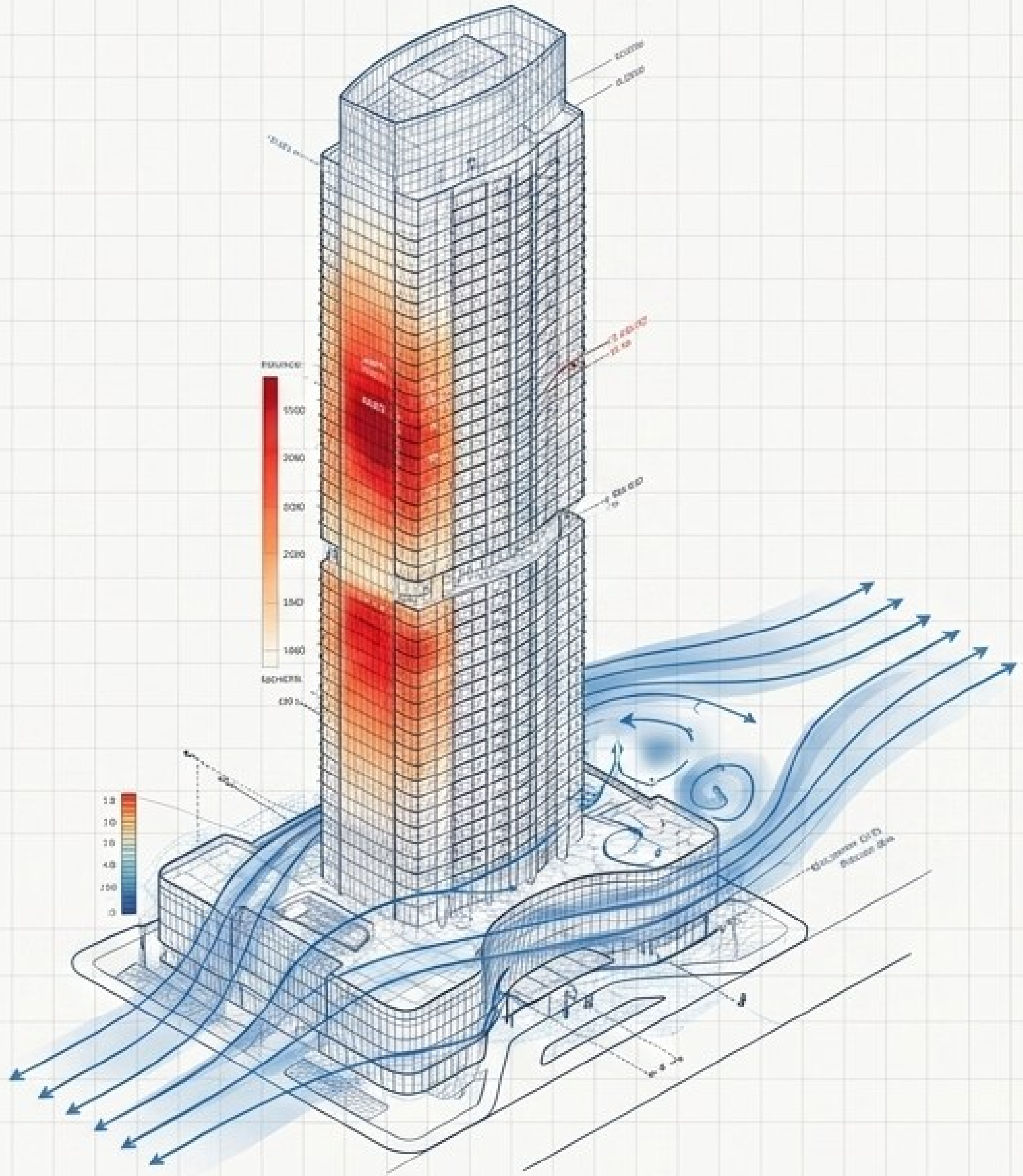
PANEL B: MASSING EVOLUTION

Translating plot boundaries, setback lines, and volume requirements into 3D massing (Physical block models, SketchUp, Rhino/Grasshopper).

Goal: Break up the massive bulk of the mixed-use tower, optimize sea views, and seamlessly integrate the 10 Townhouses and 15% Contextual Program at the podium level.



PHASE 3: THE ENVIRONMENTAL FEEDBACK LOOP



THE MANDATE:

Optimized massing must undergo rigorous environmental simulation using Ladybug/Honeybee or similar software.

SOLAR RADIATION:

The western sea-facing façade is highly vulnerable to harsh afternoon sun. Run radiation maps to identify high heat-gain zones.

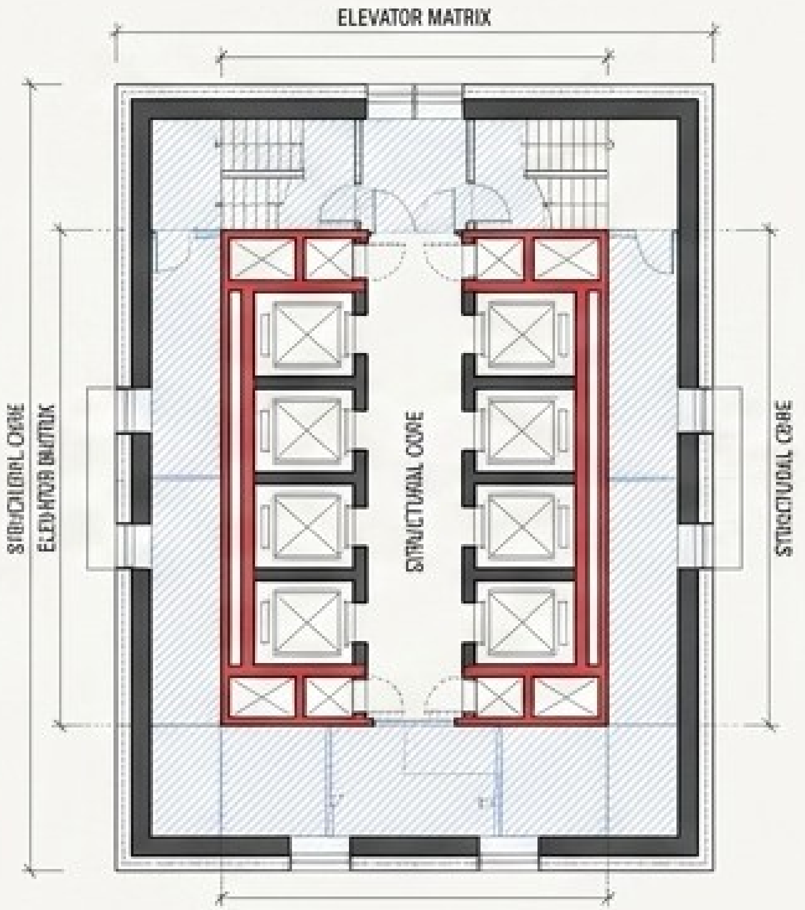
WIND ANALYSIS:

Computational Fluid Dynamics (CFD) must be run to study wind behavior. Crucial objective: Mitigate wind tunneling at pedestrian drop-offs and the clubhouse pool deck.

Phases 4 & 5: Internal Core to External Skin

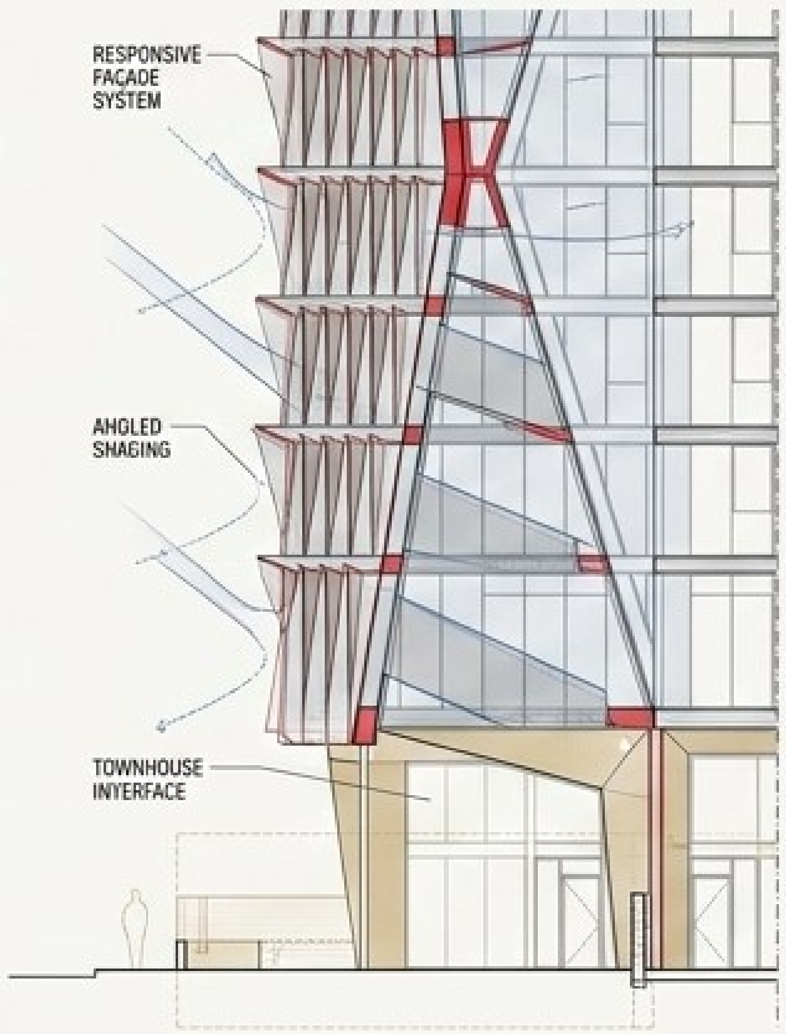
STEP 1: STRUCTURAL CORE & SPATIAL RESOLUTION (PHASE 4)

- Freeze the optimized massing shell.
- Design the central structural core, solving the intensely complex elevator matrix required to simultaneously service the the public program, 76 Rehab units, and Free Sale tower units.
- Ensure identical 1,350 sq.ft Rehab plans and calculated Free Sale units feature no structural columns disrupting primary living spaces.



STEP 2: DATA-DRIVEN ENVELOPES & DETAILING (PHASE 5)

- Translate Phase 3 heat-map data strictly into a responsive façade system (specific shading devices, varying glass specs).
- Resolve ground plane interfaces and private garden boundaries for the 10 Townhouses.



The Ultimate Standard: Deliverables & Alignment

NSQF Level 7 Alignment

Regulatory Mastery:
Independently synthesizing DCR 2034 and CRZ-II contexts.

Technical Proficiency:
Advanced massing evolution and evidence-based environmental testing.

Autonomy:
Taking full accountability for mathematical FSI proofs and extreme programmatic contradictions.

Presentation Standard (Deliverables)

- ☑ Urban & Area Matrix:
(6-layer storyboard + 3BHK/4BHK math solution).
- ☑ Iterative Massing Documentation.
- ☑ Simulation Proofs:
(Heat maps, daylight grids, CFD flows).
- ☑ Comprehensive Masterplan & Floor Plans:
(Fully furnished Free Sale, Townhouse, and Rehab typologies).
- ☑ Transverse Structural Section:
(Core, transfer girders, podium integration).